

Vision Document

1. System Overview

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Title of the project: Public Parking Management System

Objective: Design and implement a centralized software system to manage public parking lots, enabling accurate tracking of vehicle entry and exit, calculation of parking fees based on configurable tariffs, monitoring of real-time capacity, and generation of management reports for owners and administrators.

Problem: Currently, many public parking lots rely on manual, paper-based processes or simple, non-integrated tools. This leads to issues such as:

- Inaccurate or lost records of vehicle entry and exit
- Difficulty in calculating fees consistently according to different tariffs (hourly,
- Limited visibility into real-time occupancy and available spots
- Weak financial and operational reporting (revenue, peak hours, utilization)
- High dependency on operator experience and prone to human error

Target Users:

- Parking Owner: needs an overview of revenue, utilization, and operational performance.
- Parking Admin: manages configuration, tariffs, users, and access rights.
- Parking Operator: registers vehicle entry and exit, collects payments, and issues receipts.
- Customer (Driver): indirectly benefits from faster service, accurate billing, and potentially digital receipts or ticket information.

2. Stakeholders

Stakeholder | Role | Interest

Parking Owner | Business owner of one or more parking lots | Wants reliable revenue tracking, transparency, and reports on occupancy, peak hours, and financial performance to support decision-making and pricing policies.

Parking Admin | System administrator or manager assigned by the owner | Responsible for configuring the system (tariffs, users, roles, parking spots). Interested in stability, correct configuration, and easy management of users and data.

Parking Operator | Staff at the entrance/exit of the parking lot | Uses the system daily to register entry and exit, calculate fees, and issue receipts. Needs a fast, user-friendly interface with minimal errors.

Customer (Driver) | Uses the parking lot service | Wants quick entry/exit, accurate billing, and clear information (e.g., ticket, receipt, parking location).

3. Scope

In Scope:

- Management of one or more public parking lots within the system.
- Registration of vehicle entry:
- Recording vehicle plate number and entry time.
- Assigning a parking spot or area.
- Registration of vehicle exit:
- Recording exit time.
- Automatic calculation of parking fee based on defined tariffs.
- Recording payment and issuing digital or printed receipt.
- Tariff management:
- Definition and editing of tariffs by time, duration, or other rules.
- Capacity monitoring:
- Displaying total, occupied, and free spots in real time.
- User and role management:
- Creating users (admin, operator, etc.).
- Assigning roles and access rights.
- Reporting:
- Generating parking management reports (traffic, revenue, occupancy).
- Basic security (authentication and role-based authorization).

Out of Scope:

- Advanced mobile application for customers (native iOS/Android app); if needed, only basic API support will be considered.
- Integration with payment gateways for online payments (beyond the scope of initial system; assume manual payment registration).
- Integration with Automatic Number Plate Recognition (ANPR) cameras or hardware barriers (gate automation); can be considered as future work.
- Multi-city, multi-country support with complex tax rules (initial version focuses on a simple regulatory context).

4. Requirements

Functional:

- FR1: Vehicle Entry

Register vehicle entry with plate number, entry time, and optionally parking spot.

- FR2: Vehicle Exit & Fee

Register vehicle exit and automatically calculate the fee based on parking duration and active tariff.

- FR3: Tariff Management

Admin can create, edit, and delete tariff rules (hourly, daily, etc.).

- FR4: Capacity View

Show real-time total, occupied, and free spots for each parking lot.

- FR5: Receipt

Generate a receipt with plate, entry time, exit time, duration, fee, and payment status.

- FR6: User & Role Management

Manage users and assign roles (Owner, Admin, Operator) with appropriate permissions.

- FR7: Reports

Produce basic reports (number of vehicles, revenue, time period).

- FR8: Spot Management

Configure number and basic properties of parking spots.

Non-Functional:

- Performance:
Entry/exit registration and capacity view should respond within about 2 seconds for most requests.
- Scalability:
Support adding more parking lots and users without major redesign.
- Security:
User login and role-based access control; secure password storage.
- Usability:
Simple operator UI, clear error messages, and support for local language.

5. Technical Environment

Hardware:

Single server or PC for application and database; client PCs with a web browser at parking entrances/exits.

Software Stack:

Web application (e.g., Java/.NET/Node/Python) with HTML/CSS/JavaScript frontend.

Database:

Relational database (e.g., MySQL/PostgreSQL) with tables for users, roles, parking lots, spots, sessions, tariffs, and payments.

6. External Interfaces

APIs / third-party services:

None required in the first version; future integration with payment gateway or ANPR service is possible.

Data inputs/outputs:

- Inputs: vehicle plate, entry/exit time, parking spot, tariffs, user data.
- Outputs: receipts, basic reports, real-time capacity information.

7. Use Cases

- UC1 – **Register Vehicle Entry**: Operator registers vehicle arrival.
- UC2 – **Register Vehicle Exit**: Operator registers vehicle exit and receives payment.
- UC3 – **Calculate Parking Fee**: System calculates fee based on duration and tariff.
- UC4 – **Issue Receipt**: System generates a payment receipt.
- UC5 – **Define/Edit Tariffs**: Admin configures tariffs.
- UC6 – **View Parking Capacity**: Owner/Admin/Operator view current occupancy.
- UC7 – **Register Vehicle Location**: System records vehicle parking spot.
- UC8 – **Manage Users & Roles**: Owner/Admin manage users and roles.
- UC9 – **Generate Reports**: System generates simple management reports.

8. Constraints & Risks

Constraints:

Limited project time and resources; simple deployment (single server/PC); basic network environment.

Risks:

Unclear tariff rules, performance issues with large data, security misconfiguration, and UI being too complex for operators.

9. Future Work

Extensions / scalability directions:

Mobile/customer portal, ANPR and gate integration, online payments, dynamic pricing, support for many parking lots/owners, and advanced analytics.